Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Title:	**CALCULATION** **3380 ACIDULATION** **DESIGN OF FOUNDATION**
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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to prepare design of foundation for **Acidulation Reactor (3380-REA-001,002) & T-40 Acidulation Thickener (3380-THK-001)**. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from Equipment load data drawings & designed in excel sheet.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Load Data drawings attached in Appendix A. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations

Foundation shall be Earth Foundation with a Concrete Ring wall for Tanks. The design calculations are prepared as per recommendation of standard API 650 - Annex B. Foundation of thickener are of isolated type.

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5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	75
Exposed to weather or contact with ground	Column,Wall	50

5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 350 Kn/m^2 at 1.25 m below EGL for the tanks & 400 Kn/m^2 at 1.5m below E.G.L for the Thickener foundations. (Refer Table 5 - BH-8 & BH-9 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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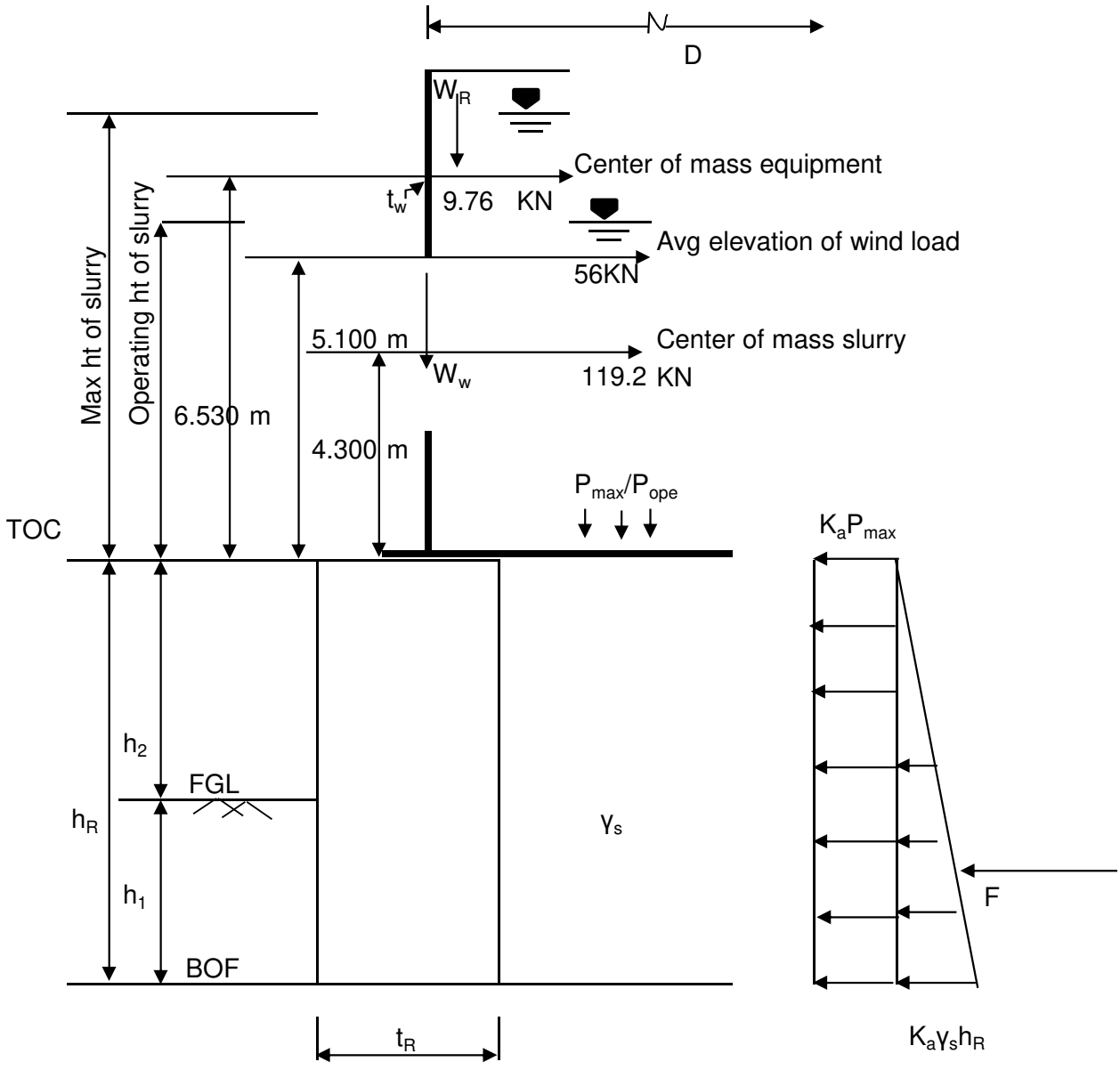
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6. DESIGN CALCULATION FOR FOUNDATION

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Design Calculation of Foundation for Acidulation Reactor



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	6.5	m
Self weight of tank =	228	KN
Self weight of tank per m of circumference,W_w =	11.17	KN/m
External platform weight =	83	KN
Self weight of tank roof per m of circumference,W_R=	4.065	KN/m
Maximum weight of slury=	3070	KN
Pressure at base of tank ,P_{max}	92.52	KN/m^2
Operating weight of slury=	2785	KN
Pressure at base of tank ,P_{ope}	83.93	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete,γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil,γ_s =	19	Kn/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	350	Kn/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	1.95	m
FGL to Bottom of foundation ,h_1 =	1.25	m
FGL to Top of concrete ,h_2 =	0.7	m
Ringwall thickness, t_R =	450	mm
Inner dia of ringwall,D_i =	6.05	m
Outer dia of ringwall,D_o =	6.95	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	14.03	m^3

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Check of soil bearing pressure (Max Slurry Load): For wind Load

Wind force = 56 kn acting at a height of 7.050 m from BOF

M_{OT} = Overturning moment = 394.8 KN-m

M_{OT} / S_{xx} = 28.1 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 127 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 155 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 98.8 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = (W_w + W_R)/t_R + h_R\gamma_c$ = 80.6 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 108.8 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 52.51 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 394.8 KN-m

M_{OR} = Restoring moment = (228+PI()*6.5*0.45*1.95*25)*6.5/2 = 2196.9 KN-m

FOS = M_{OR} / M_{OT} 5.56 > 1.5 ok

Check for soil bearing pressure (Operating Slurry Load): For Seismic Load

Seismic force at center of mass equipment = 228/(228+2785)*129 = 9.762 KN

acting at a height of 8.480 m from BOF

Seismic force at center of mass slurry = 2785/(228+2785)*129 = 119.2 KN

acting at a height of 6.250 m from BOF

M_{OT} = Overturning moment = 828 KN-m

M_{OT} / S_{xx} = 59 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{ope}/2 + h_R\gamma_c$ = 123 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 164 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 81.3 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 126.9 Kn/m^2 < 350.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 129.6 Kn/m^2 < 350.0 KN/m^2 Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 71.46 Kn/m

Hoop tension in ringwall, T = 1/2FD = 232 Kn

Factored design tension, T_u= 325 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 860.1 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 1755 mm^2

$A_{s,reqd}$ (per face of ring wall)= 877.5 mm^2

Required : 12 dia bar @ 251.5 c/c (per face)

Provide : 12 dia bar @ 200 c/c (per face)

$A_{s,provided}$(per face) = 1103 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 540 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 270 mm^2

Required : 10 dia bar @ 291.1 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{s,provided}$(per face) = 392.9 mm^2 ok

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DESIGN OF FOUNDATION MKD. F1

	Along-X	Along-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	1.80m	1.80m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.70 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

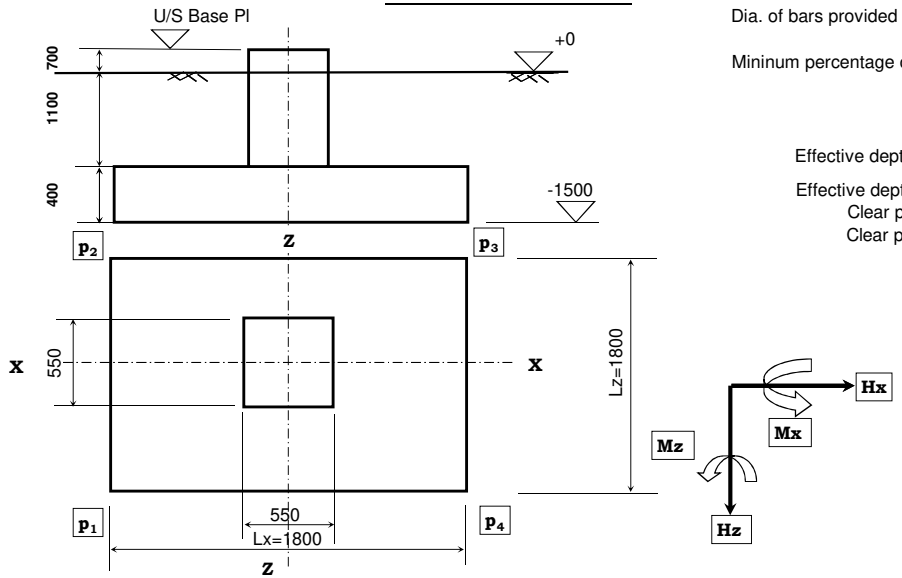
Tank Dead Load	D
Max. Fluid Load	F1
Normal Operational Fluid Load	F2
Wind Load	W
Earthquake Load	E

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Mininum percentage of reinforcement at bottom=	0.18%

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.8*1.8*0.4*24)+(0.55*0.55*1.8*24) =	44.17KN
Weight of soil = ((1.8*1.8)-(0.55*0.55))*1.1*18 =	58.16KN
Total =	102.33KN



Effective depth of slab along X direction = 317mm = 0.317m

Effective depth of slab along Z direction = 301mm = 0.301m

Clear projection along Z direction = 0.625m

Clear projection along X direction = 0.625m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F1

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mz (kNm)
1	Tank Dead Load (D)	106	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	303	47.0	0.0	0.0	0.0
3	Normal Operational Fluid Load (F2)	244	38.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	11.0	5.0	0.0	0.0	0.0
5	Earthquake Load (E)(+/-)	10.0	7.0	0.0	0.0	0.0

(Bridge/Drive loads are added with dead load to be on the conservative side)

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	D + F1	409.0	47.0	0.0	0.0	0.0	511.33	47.00	103.40	0.00	0.00
202	D + F1 + W	420.0	52.0	0.0	0.0	0.0	522.33	52.00	114.40	0.00	0.00
203	D + F2+ 0.7 E	357.0	42.9	0.0	0.0	0.0	459.33	42.90	94.38	0.00	0.00
204	0.6 D + W	74.6	5.0	0.0	0.0	0.0	176.93	5.00	11.00	0.00	0.00
205	0.6(D+F2)+0.7 E	217.0	27.7	0.0	0.0	0.0	319.33	27.70	60.94	0.00	0.00
Ultimate Conditions											
301	1.4 (D + F1)	572.6	65.8	0.0	0.0	0.0	715.86	65.80	144.76	0.00	0.00
302	1.2(D + F1) + W	501.8	61.4	0.0	0.0	0.0	624.60	61.40	135.08	0.00	0.00
303	1.2(D + F2) + E	500.8	63.4	0.0	0.0	0.0	623.60	63.40	139.48	0.00	0.00
304	0.9 D + 1.6W	113.0	8.0	0.0	0.0	0.0	205.10	8.00	17.60	0.00	0.00
305	0.9 (D+ F2) + E	325.0	41.2	0.0	0.0	0.0	417.10	41.20	90.64	0.00	0.00

Check for overturning and sliding:

	Load Cases:	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D + F1	511.33	47.00	103.40	0.00	0.00				
202	D + F1 + W	522.33	52.00	114.40	0.00	0.00	4.11		4.02	
203	D + F2+ 0.7 E	459.33	42.90	94.38	0.00	0.00	4.38		4.28	
204	0.6 D + W	176.93	5.00	11.00	0.00	0.00	14.48		14.15	
205	0.6(D+F2)+0.7 E	319.33	27.70	60.94	0.00	0.00	4.72		4.61	

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DESIGN OF FOUNDATION MKD. F1

Load Combination & Calculation of Base pressure

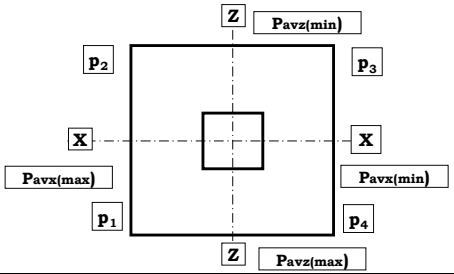
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 427 kN/m^2 For WL/SL case = 527 kN/m^2



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	D + F1	511.33	103.40	0.00	157.82	106.38	0.00	264.20	264.20	51.44	51.44	237.20	264.20	51.44	157.82	157.82	O.K
202	D + F1 + W	522.33	114.40	0.00	161.21	117.70	0.00	278.91	278.91	43.52	43.52	251.91	278.91	43.52	161.22	161.22	O.K
203	D + F2+ 0.7 E	459.33	94.38	0.00	141.77	97.10	0.00	238.87	238.87	44.67	44.67	211.87	238.87	44.67	141.77	141.77	O.K
204	0.6 D + W	176.93	11.00	0.00	54.61	11.32	0.00	65.92	65.92	43.29	43.29	38.92	65.92	43.29	54.61	54.61	O.K
205	0.6(D+F2)+0.7 E	319.33	60.94	0.00	98.56	62.70	0.00	161.25	161.25	35.86	35.86	134.25	161.25	35.86	98.56	98.56	O.K
301	1.4 (D + F1)	715.86	144.76	0.00	220.95	148.93	0.00	369.88	369.88	72.02	72.02	342.88	369.88	72.02	220.95	220.95	
302	1.2(D + F1) + W	624.60	135.08	0.00	192.78	138.97	0.00	331.75	331.75	53.81	53.81	304.75	331.75	53.81	192.78	192.78	
303	1.2(D + F2) + E	623.60	139.48	0.00	192.47	143.50	0.00	335.97	335.97	48.97	48.97	308.97	335.97	48.97	192.47	192.47	
304	0.9 D + 1.6W	205.10	17.60	0.00	63.30	18.11	0.00	81.41	81.41	45.19	45.19	54.41	81.41	45.19	63.30	63.30	
305	0.9 (D+ F2) + E	417.10	90.64	0.00	128.73	93.25	0.00	221.98	221.98	35.48	35.48	194.98	221.98	35.48	128.73	128.73	
	Max :	716	145	0													
	Min :	177	11	0													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 237.20 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 257.20 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 342.88 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 362.88 kN/m^2

Sample calculation of values of base pressure:

Load case : D + F1 + W

Maximum pressure, p_1 (pmax) = 161.21+117.70+0.00= 278.91 kN/m^2

p_2= 161.21+117.70-0.00= 278.91 kN/m^2

p_3= 161.21-(117.70+0.00)= 43.52 kN/m^2

p_4= 161.21-117.70+0.00= 43.52 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 278.91-1.5*18= 251.91 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (278.91+278.91)/2= 278.91 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (43.52+43.52)/2= 43.52 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (278.91+43.52)/2= 161.21 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (43.52+278.91)/2= 161.21 kN/m^2

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DESIGN OF FOUNDATION MKD. F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.55+2*0.5*0.301)+(0.55+2*0.5*0.301))*2

= 3.404 m

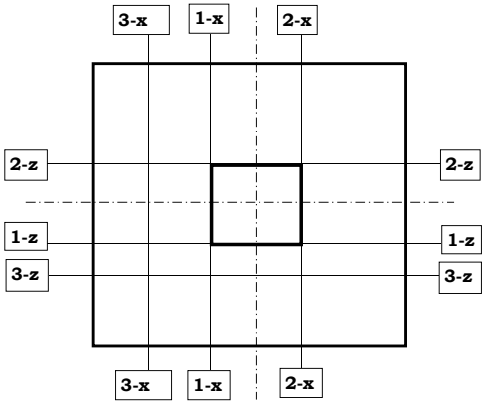
Punching Area= ((0.55+2*0.5*0.301)*(0.55+2*0.5*0.301))

= 0.72 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.1*18)

= 29.40 kN/m^2

Area of footing, A_f= 3.24 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M$_{1x}$ kNm	B.M$_{2x}$ kNm	S.F$_{3x}$ kN	S.F$_{(punching)}$ kN
Ultimate Condition								
301	1.4 (D + F1)	266.46	175.44	318.91	59.77	15.06	97.02	482.69
302	1.2(D + F1) + W	235.24	150.32	284.19	52.77	11.05	85.80	411.71
303	1.2(D + F2) + E	236.32	148.62	286.86	53.39	10.31	86.86	410.93
304	0.9 D + 1.6W	68.83	57.77	75.21	9.34	3.90	15.06	85.43
305	0.9 (D+ F2) + E	157.22	100.24	190.07	33.40	5.40	54.40	250.32

Sample calculation of BM & SF

Load case : 301

p_{1x} = 369.88-(369.88-72.02)*0.625/1.8 = 266.46 kN/m^2 Overhang = 0.625 m

p_{2x} = 72.02+(369.88-72.02)*0.625/1.8 = 175.44 kN/m^2 Effective depth = 0.317 m

p_{3x} = 369.88-(369.88-72.02)*(0.625-0.317)/1.8 = 318.91 kN/m^2

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BM_{1x} = (266.46-29.4)*0.625*0.625/2+0.5*(369.88-266.46)*0.625*0.625*0.67 = 59.77 kN-m/m

BM_{2x} = (175.44-29.4)*0.625*0.625/2-0.5*(-72.02+175.44)*0.625*0.625*0.67 = 15.06 kN-m/m

SF_{3x} = (318.91-29.4)*(0.625-0.317)+0.5*(369.88-318.91)*(0.625-0.317) = 97.02 kN/m

$S.F_{(punching)}$ = (3.24-0.72)*(220.95-29.4) = 482.69 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	59.77	0.59	0.79	0.002	606.73	570.60	16ϕT	331.39	300	200	1005.31
At top	0.00	0.00	0.00	0.000	0.00	190.20	10ϕT	412.93	300	200	392.70

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 97.02 kN/m

Shear stress , tv= (97.02*1000)/(1000*317)

= 0.31 N/mm2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 482.69 kN

Strength of Concrete in shear Vc1 = 3191.93 kN Table 22.6.5.2_ (a)

Strength of Concrete in shear Vc2 = 2054.81 kN Table 22.6.5.2_ (b)

Strength of Concrete in shear Vc3 = 2127.95 kN Table 22.6.5.2_ (c)

Design Vc= 2054.81 kN

Allowable shear stress, $(\phi\ Vc)/bd$ = 1.43 N/mm2

Design punching shear stress, τ = (482.69*1000)/(3404*317)

= 0.45 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p_{1z} =pressure at the face of column at section 1z-1z in kN/m^2.
p_{2z} =pressure at the face of column at section 2z-2z in kN/m^2.
p_{3z} =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M$_{1z}$ =Moment at the face of column at section 1z-1z in kN-m/m.
B.M$_{2z}$ =Moment at the face of column at section 2z-2z in kN-m/m.
S.F $_{3z}$ = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	B.M$_{1z}$ kNm	B.M$_{2z}$ kNm	S.F$_{3z}$ kN
Ultimate Conditions							
301	1.4 (D + F1)	220.95	220.95	220.95	37.41	37.41	62.06
302	1.2(D + F1) + W	192.78	192.78	192.78	31.91	31.91	52.94
303	1.2(D + F2) + E	192.47	192.47	192.47	31.85	31.85	52.83
304	0.9 D + 1.6W	63.30	63.30	63.30	6.62	6.62	10.98
305	0.9 (D+ F2) + E	128.73	128.73	128.73	19.40	19.40	32.18

Sample calculation of BM & SF

Load case : 301

p_{1z} = 220.95-(220.95-220.95)*0.625/1.8= 220.95 kN/m^2

p_{2z} = 220.95+(220.95-220.95)*0.625/1.8= 220.95 kN/m^2 Overhang = 0.625 m

p_{3z} = 220.95-(220.95-220.95)*(0.625-0.301)/1.8= 220.95 kN/m^2 Effective depth = 0.301 m

BM_{1z} = (220.95-29.4)*0.625*0.625/2+0.5*(220.95-220.95)*0.625*0.625*0.67 = 37.41 kN-m/m

BM_{2z} = (220.95-29.4)*0.625*0.625/2-0.5*(-220.95+220.95)*0.625*0.625*0.67 = 37.41 kN-m/m

SF_{3z} = (220.95-29.4)*(0.625-0.301)+0.5*(220.95-220.95)*(0.625-0.301) = 62.06 kN/m

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	37.41	0.413	0.551	0.0013	419.47	541.80	16ϕT	371.10	300	200	1005.3
At top	0.00	0.000	0.000	0.0000	0.00	180.60	10ϕT	434.88	300	200	392.7

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm2

Shear at a distance 'd' from face of column, V_u= 62.06 kN/m

Shear stress , τ_v= (62.06*1000)/(1000*301)

= 0.21 N/mm2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	10ϕT	200	c/c
Along Z Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	10ϕT	200	c/c

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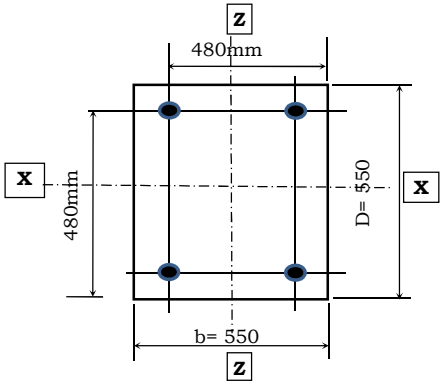
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DESIGN OF COLUMN MKD C1

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	20mm
Dia. of link=	10mm
Effective cover =	70mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%

Table 20.6.1.3.1 of SBC-304



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	1800mm
Unsupproted length of column about Z (l_{uz}) =	1800mm

For Columns not braced against sideway,

Where, Effective length factor K= 2

Slenderness ratio, $K\,l_{ux} / r_x$ = 21.82 < 22 SECTION 6.2.5a

Radius of gyration r_x= 165 mm

$K\,l_{uz} / r_z$ = 21.82 < 22 SECTION 6.2.5a

Radius of gyration r_z= 165 mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	106	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	303	47.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	244	38.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	11.0	5.0	0.0	0.0	0.0
5	Earthquake Load (E)(+/-)	10.0	7.0	0.0	0.0	0.0
			γ_x =	0.78		
			γ_z =	0.78		

(Bridge/Drive loads are added with dead load to be on the conservative side)

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Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	57.26	11.8	0.0	88.1	18.2	0.0	0.08
302	1.2(D + F1) + W	50.2	11.1	0.0	77.2	17.0	0.0	0.07
303	1.2(D + F2) + E	50.1	9.5	0.0	77.0	14.6	0.0	0.07
304	0.9 D + 1.6W	11.3	0.9	0.0	17.4	1.4	0.0	0.02
305	0.9 (D+ F2) + E	32.5	7.4	0.0	50.0	11.4	0.0	0.05

Reinforcement: Provide,

4-Nos	20 mm Dia
4-Nos	20 mm Dia
4-Nos	20 mm Dia

1257.14 Sq mm (At Corner)
1257.14 Sq mm (Along Z axis)
1257.14 Sq mm (Along Y axis)

Asc pvd. = 3771.43 Sq mm

Total Asc = 1.2 %

β = P_n / P_0 (range is 0.55 to 0.70)

Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1047.1175 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
301	0.08	0.08	46.6	46.6	0.55	1.16	0.34	O.K
302	0.08	0.08	46.6	46.6	0.55	1.16	0.31	O.K
303	0.08	0.08	46.6	46.6	0.55	1.16	0.26	O.K
304	0.06	0.06	34.9	34.9	0.55	1.16	0.02	O.K
305	0.06	0.06	34.9	34.9	0.55	1.16	0.27	O.K

Links, 10mm dia @ 300 C/C

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DESIGN OF FOUNDATION MKD. F2

	Along-X	Along-Z
Size of column :	0.65m	0.65m
Size of fdn. raft :	3.35m	3.35m
Thickness of raft :	0.65 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.70 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Tank Dead Load	D
Max. Fluid Load	F1
Normal Operational Fluid Load	F2
Wind Load	W
Earthquake Load	E

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.06%

Calculation of Weight of Foundation :

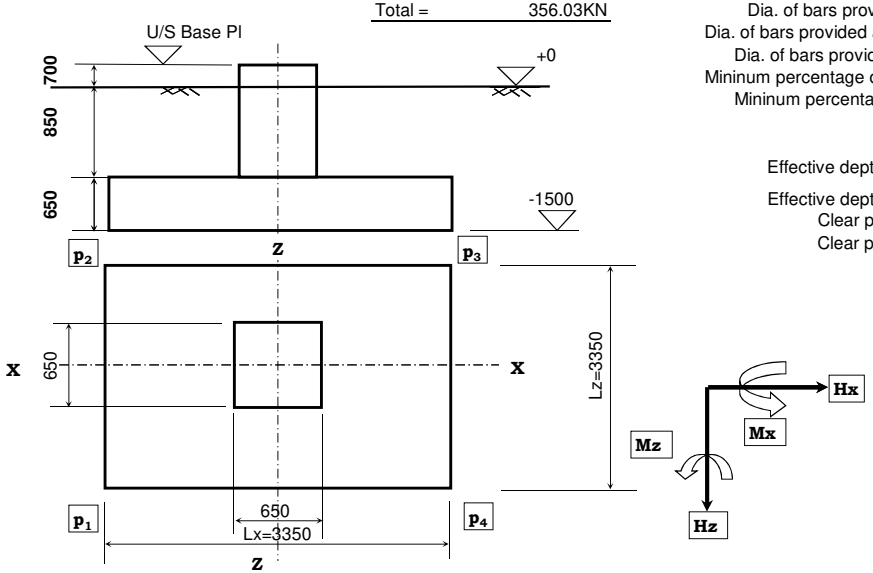
Weight of raft & Col.=	(3.35*3.35*0.65*24)+(0.65*0.65*1.55*24) =	190.79KN
Weight of soil =	((3.35*3.35)-(0.65*0.65))*0.85*18 =	165.24KN
	Total =	356.03KN

Effective depth of slab along X direction = 565mm = 0.565m

Effective depth of slab along Z direction = 545mm = 0.545m

Clear projection along Z direction = 1.35m

Clear projection along X direction = 1.35m



H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F2

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	40	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	2700	0.0	0.0	0.0	0.0
3	Normal Operational Fluid Load (F2)	2172	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	0.0	7.0	0.0	7.0	0.0
5	Earthquake Load (E)(+/-)	0.0	10.0	0.0	9.0	0.0

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	D + F1	2740.0	0.0	0.0	0.0	0.0	3096.03	0.00	0.00	0.00	0.00
202	D + F1 + W	2740.0	7.0	0.0	0.0	0.0	3096.03	7.00	15.40	0.00	0.00
203	D + F2+ 0.7 E	2212.0	7.0	0.0	0.0	0.0	2568.03	7.00	15.40	0.00	0.00
204	0.6 D + W	24.0	7.0	7.0	0.0	0.0	380.03	7.00	22.40	0.00	0.00
205	0.6(D+F2)+0.7 E	1327.2	7.0	6.3	0.0	0.0	1683.23	7.00	21.70	0.00	0.00
Ultimate Conditions											
301	1.4 (D + F1)	3836.0	0.0	0.0	0.0	0.0	4334.44	0.00	0.00	0.00	0.00
302	1.2(D + F1) + W	3288.0	7.0	0.0	0.0	0.0	3715.24	7.00	15.40	0.00	0.00
303	1.2(D + F2) + E	3288.0	10.0	0.0	0.0	0.0	3715.24	10.00	22.00	0.00	0.00
304	0.9 D + 1.6 W	36.0	11.2	11.2	0.0	0.0	356.43	11.20	35.84	0.00	0.00
305	0.9 (D+ F2) + E	1990.8	10.0	9.0	0.0	0.0	2311.23	10.00	31.00	0.00	0.00

Check for overturning and sliding:

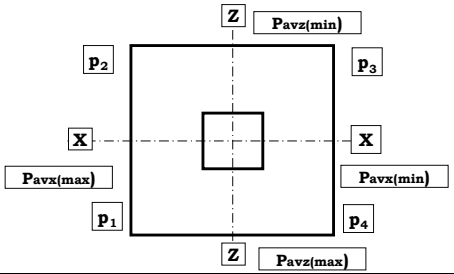
	Load Cases:	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D + F1	3096.03	0.00	0.00	0.00	0.00				
202	D + F1 + W	3096.03	7.00	15.40	0.00	0.00	336.74		176.92	
203	D + F2+ 0.7 E	2568.03	7.00	15.40	0.00	0.00	279.31		146.74	
204	0.6 D + W	380.03	7.00	22.40	0.00	0.00	28.42		21.72	
205	0.6(D+F2)+0.7 E	1683.23	7.00	21.70	0.00	0.00	129.93		96.18	

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DESIGN OF FOUNDATION MKD. F2



Load Combination & Calculation of Base pressure

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 427 kN/m^2 For WL/SL case = 527 kN/m^2

Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	D + F1	3096.03	0.00	0.00	275.88	0.00	0.00	275.88	275.88	275.88	275.88	248.88	275.88	275.88	275.88	275.88	O.K
202	D + F1 + W	3096.03	15.40	0.00	275.88	2.46	0.00	278.33	278.33	273.42	273.42	251.33	278.33	273.42	275.88	275.88	O.K
203	D + F2+ 0.7 E	2568.03	15.40	0.00	228.83	2.46	0.00	231.29	231.29	226.37	226.37	204.29	231.29	226.37	228.83	228.83	O.K
204	0.6 D + W	380.03	22.40	0.00	33.86	3.57	0.00	37.44	37.44	30.29	30.29	10.44	37.44	30.29	33.87	33.87	O.K
205	0.6(D+F2)+0.7 E	1683.23	21.70	0.00	149.99	3.46	0.00	153.45	153.45	146.52	146.52	126.45	153.45	146.52	149.99	149.99	O.K
301	1.4 (D + F1)	4334.44	0.00	0.00	386.23	0.00	0.00	386.23	386.23	386.23	386.23	359.23	386.23	386.23	386.23	386.23	
302	1.2(D + F1) + W	3715.24	15.40	0.00	331.05	2.46	0.00	333.51	333.51	328.59	328.59	306.51	333.51	328.59	331.05	331.05	
303	1.2(D + F2) + E	3715.24	22.00	0.00	331.05	3.51	0.00	334.56	334.56	327.54	327.54	307.56	334.56	327.54	331.05	331.05	
304	0.9 D + 1.6 W	356.43	35.84	0.00	31.76	5.72	0.00	37.48	37.48	26.04	26.04	10.48	37.48	26.04	31.76	31.76	
305	0.9 (D+ F2) + E	2311.23	31.00	0.00	205.95	4.95	0.00	210.89	210.89	201.00	201.00	183.89	210.89	201.00	205.95	205.95	
	Max :	4334	36	0													
	Min :	356	0	0													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 248.88 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 268.88 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 359.23 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 379.23 kN/m^2

Sample calculation of values of base pressure:

Load case : D + F1 + W

Maximum pressure, p_1 (pmax) = 275.88+2.46+0.00= 278.33 kN/m^2

p_2= 275.88+2.46-0.00= 278.33 kN/m^2

p_3= 275.88-(2.46+0.00)= 273.42 kN/m^2

p_4= 275.88-2.46+0.00= 273.42 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 278.33-1.5*18= 251.33 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (278.33+278.33)/2= 278.33 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (273.42+273.42)/2= 273.42 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (278.33+273.42)/2= 275.88 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (273.42+278.33)/2= 275.88 kN/m^2

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DESIGN OF FOUNDATION MKD. F2

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.65+2*0.5*0.545)+(0.65+2*0.5*0.545))*2

= 4.78 m

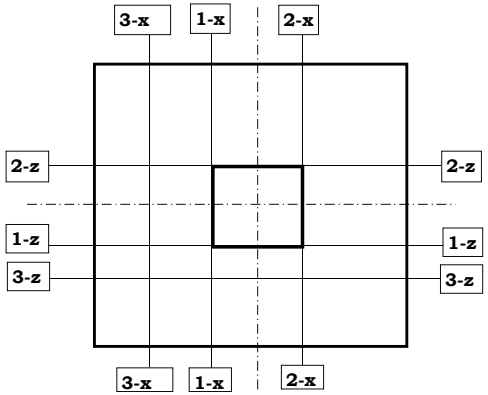
Punching Area= ((0.65+2*0.5*0.545)*(0.65+2*0.5*0.545))

= 1.43 m2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.65*24+0.85*18)

= 30.90 kN/m2

Area of footing, Af= 11.22 m2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	S.F(punching) kN
Ultimate Conditions								
301	1.4 (D + F1)	386.23	386.23	386.23	323.79	323.79	278.93	3479.55
302	1.2(D + F1) + W	331.53	330.57	332.36	275.15	271.87	237.10	2939.24
303	1.2(D + F2) + E	331.73	330.37	332.92	275.85	271.17	237.73	2939.24
304	0.9 D + 1.6 W	32.87	30.65	34.80	4.60	-3.03	4.11	8.42
305	0.9 (D+ F2) + E	206.90	204.99	208.57	162.81	156.21	140.38	1714.14

Sample calculation of BM & SF

Load case : 301

p_{1x} = 386.23-(386.23-386.23)*1.35/3.35 = 386.23 kN/m^2

p_{2x} = 386.23+(386.23-386.23)*1.35/3.35 = 386.23 kN/m^2

p_{3x} = 386.23-(386.23-386.23)*(1.35-0.565)/3.35 = 386.23 kN/m^2

Overhang = 1.35 m

Effective depth = 0.565 m

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BM_{1x} = (386.23-30.9)*1.35*1.35/2+0.5*(386.23-386.23)*1.35*1.35*0.67 = 323.79 kN-m/m

BM_{2x} = (386.23-30.9)*1.35*1.35/2-0.5*(-386.23+386.23)*1.35*1.35*0.67 = 323.79 kN-m/m

SF_{3x} = (386.23-30.9)*(1.35-0.565)+0.5*(386.23-386.23)*(1.35-0.565) = 278.93 kN/m

$S.F_{(punching)}$ = (11.2225-1.43)*(386.23-30.9) = 3479.55 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	323.79	1.01	1.35	0.003	1862.67	1017.00	20ϕT	168.66	300	150	2094.40
At top	-3.03	-0.01	0.01	0.000	17.02	339.00	10ϕT	231.68	300	150	523.60

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi Vc)/bd$ = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 278.93 kN/m

Shear stress , τ_v= (278.93*1000)/(1000*565)

= 0.49 N/mm2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 3479.55 kN

Strength of Concrete in shear Vc1 = 7988.78 kN Table 22.6.5.2_ (a)

Strength of Concrete in shear Vc2 = 5810.53 kN Table 22.6.5.2_ (b)

Strength of Concrete in shear Vc3 = 5325.85 kN Table 22.6.5.2_ (c)

Design Vc= 5325.85 kN

Allowable shear stress, $(\phi Vc)/bd$ = 1.48 N/mm2

Design punching shear stress, τ = (3479.55*1000)/(4780*565)

= 1.29 N/mm2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p_{1z} =pressure at the face of column at section 1z-1z in kN/m2.
p_{2z} =pressure at the face of column at section 2z-2z in kN/m2.
p_{3z} =pressure at a distance d from the face of column at section 3z-3z in kN/m2.
$B.M_{1z}$ =Moment at the face of column at section 1z-1z in kN-m/m.
$B.M_{2z}$ =Moment at the face of column at section 2z-2z in kN-m/m.
$S.F_{3z}$ = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
301	1.4 (D + F1)	386.23	386.23	386.23	323.79	323.79	286.04
302	1.2(D + F1) + W	331.05	331.05	331.05	273.51	273.51	241.62
303	1.2(D + F2) + E	331.05	331.05	331.05	273.51	273.51	241.62
304	0.9 D + 1.6 W	31.76	31.76	31.76	0.78	0.78	0.69
305	0.9 (D+ F2) + E	205.95	205.95	205.95	159.51	159.51	140.92

Sample calculation of BM & SF

Load case : 301

p_{1z} = 386.23-(386.23-386.23)*1.35/3.35= 386.23 kN/m2

p_{2z} = 386.23+(386.23-386.23)*1.35/3.35= 386.23 kN/m2

p_{3z} = 386.23-(386.23-386.23)*(1.35-0.545)/3.35= 386.23 kN/m2

Overhang = 1.35 m

Effective depth = 0.545 m

BM_{1z} = (386.23-30.9)*1.35*1.35/2+0.5*(386.23-386.23)*1.35*1.35*0.67 = 323.79 kN-m/m

BM_{2z} = (386.23-30.9)*1.35*1.35/2-0.5*(-386.23+386.23)*1.35*1.35*0.67 = 323.79 kN-m/m

SF_{3z} = (386.23-30.9)*(1.35-0.545)+0.5*(386.23-386.23)*(1.35-0.545) = 286.04 kN/m

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm2)	ρ=	Ast (mm2)	Minm. Ast (mm2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm2)
At bottom	323.79	1.090	1.454	0.0035	2005.56	981.00	20ϕT	156.64	300	150	2094.4
At top	0.00	0.000	0.000	0.0000	0.00	327.00	10ϕT	240.18	300	150	523.6

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, V_u= 286.04 kN/m

Shear stress , τ_v= (286.04*1000)/(1000*545)

= 0.52 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	20ϕT	150	c/c
	At Top	10ϕT	150	c/c
Along Z Direction :				
	At Bottom	20ϕT	150	c/c
	At Top	10ϕT	150	c/c

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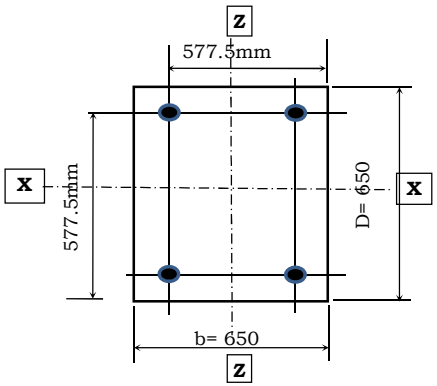
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DESIGN OF COLUMN MKD C2

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	72.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%

Table 20.6.1.3.1 of SBC-304



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	1600mm
Unsupproted length of column about Z (l_{uz}) =	1600mm

For Columns not braced against sideway,

Where, Effective length factor K= 2

Slenderness ratio, $K\,l_{ux} / r_x$ = 16.41 < 22 SECTION 6.2.5a

Radius of gyration r_x= 195 mm

$K\,l_{uz} / r_z$ = 16.41 < 22 SECTION 6.2.5a

Radius of gyration r_z= 195 mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	40	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	2700	0.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	2172	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	0.0	7.0	0.0	7.0	0.0
5	Earthquake Load (E)(+/-)	0.0	10.0	0.0	9.0	0.0
			γ_x =	0.81		
			γ_z =	0.81		

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Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	383.60	0.0	0.0	590.2	0.0	0.0	0.40
302	1.2(D + F1) + W	328.8	1.8	0.0	505.8	2.8	0.0	0.34
303	1.2(D + F2) + E	328.8	2.5	0.0	505.8	3.8	0.0	0.34
304	0.9 D + W	3.6	1.8	0.0	5.5	2.8	0.0	0.00
305	0.9 (D+ F2) + E	199.1	2.5	0.0	306.3	3.8	0.0	0.21

Reinforcement: Provide,

4-Nos	25 mm Dia	1964.29 Sq mm	(At Corner)	Asc pvd. = 4478.57 Sq mm
4-Nos	20 mm Dia	1257.14 Sq mm	(Along Z axis)	
4-Nos	20 mm Dia	1257.14 Sq mm	(Along Y axis)	Total Asc = 1.1 %

β = Pn / P0 (range is 0.55 to 0.70) Where, P0 = 0.85 f 'c (Ag - Asc) + fy Asc = 1431.7138 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cb^2D$						
301	0.12	0.12	115.3	115.3	0.55	1.16	0.00	O.K
302	0.12	0.12	115.3	115.3	0.55	1.16	0.01	O.K
303	0.12	0.12	115.3	115.3	0.55	1.16	0.02	O.K
304	0.05	0.05	48.1	48.1	0.55	1.16	0.04	O.K
305	0.11	0.11	105.7	105.7	0.55	1.16	0.02	O.K

Links, 10mm dia @ 300 C/C

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

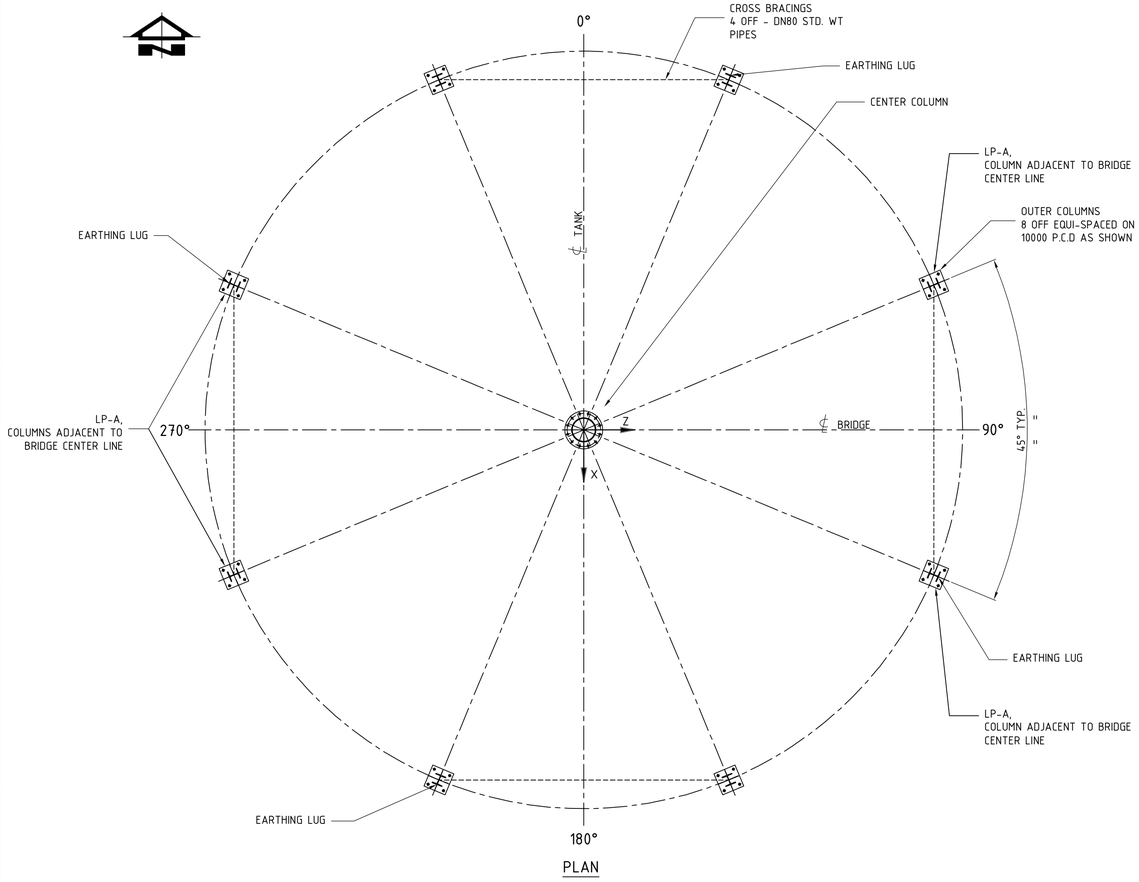
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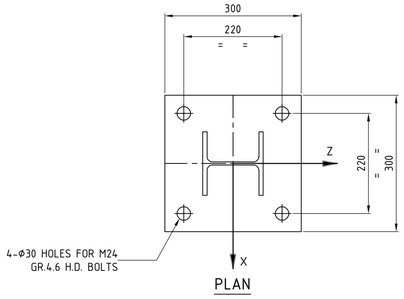
ATTACHMENT 1: LOAD DATA DRAWINGS

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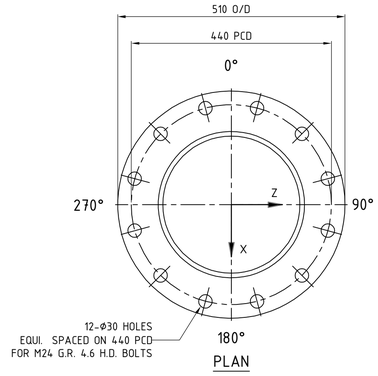
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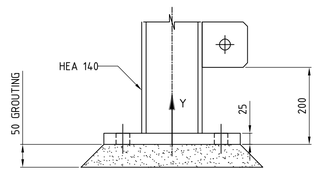
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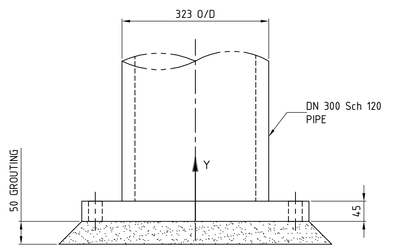
PLAN



PLAN



ELEVATION
OUTER COLUMN
SCALE 1:6



ELEVATION
CENTRE COLUMN
SCALE 1:6

LOAD TABLE

	Column PCD (mm)	Tank Dead Load	Max Fluid Load SG 1.708		Normal Operating Fluid Load SG 1.374		Seismic Load (+/-)			Wind Load (+/-)			Bridge/Drive Dead Load (LP-A) *NOTE 9	Bridge/Drive Live Load (LP-A) *NOTE 9
		Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Vertical (Y) (kN)	Vertical (Y) (kN)
Outer Columns	10 000	13	47	303	38	244	7	10	0	5	11	0	25	68
Central Column	0	40	0	2 700	0	2 172	10	0	9	7	0	7	0	0
Total Applied Load	–	140	0	5 124	0	4 122	40	0	136	25	0	99	73	170

NOTES:

1. ALL LOADS LISTED ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301
2. SEISMIC LOADS ARE CALCULATED CONSIDERING DYNAMIC SLOSHING EFFECTS OF NORMAL OPERATING SG LIQUIDS
3. COMBINATIONS INCLUDING SEISMIC LOAD TO USE NORMAL OPERATING FLUID LOAD
4. ALL HORIZONTAL REACTIONS ARE IN THE PLANE OF BRACING
5. TOTAL LOADS AND MOMENTS ARE CALCULATED AT THE UNDERSIDE OF BASEPLATE
6. COLUMN LOADS ARE GIVEN PER COLUMN
7. DIFFERENTIAL SETTLEMENT BETWEEN ANY ADJACENT COLUMN SHALL BE LIMITED TO L / 600
8. TILTING OF THE TANK DUE TO OUTER COLUMN DIFFERENTIAL SETTLEMENT SHALL BE LIMITED TO 8 mm
9. BRIDGE/DRIVE LOADS REACTED AT LOAD POINTS A (LP-A) ONLY. LP-A IS ADJACENT TO BRIDGE CENTER LINE.
10. LENGTH OF ANCHOR BOLTS TO BE DEFINED AS A FUNCTION OF CIVIL ENGINEERING

SEISMIC DATA TO SBC301

Ss, S1:	0.10, 0.02
Importance Factor:	1
Site Class:	B

WIND DATA TO SBC301

V:	43 m/s
Exposure Class:	C

FOUNDATION TOLERANCES

HEIGHT OF COLUMN FOUNDATIONS	± 10 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT IN A GROUP	± 3 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT GROUP	± 6 mm
ANCHOR BOLTS - MAX TOLERANCE ACCUMULATION BETWEEN BOLT GROUPS	± 6 mm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-ME-DED-1001	GENERAL NOTES								
MD-216-3000-EG-ME-SPC-3001	TECHNICAL SPECIFICATION								
MD-216-3380-EG-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3380-EG-GAD-1001	EQUIPMENT GENERAL ARRANGEMENT DRAWING	A01	14.11.2019	ISSUED FOR APPROVAL	TK	AC	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-D-03335-001 — REVISION: 0

OUTOTEC DRAWING NUMBER: 900707 __GAU01 __HL15 __MLD15 __33801 — DOCUMENT ID: OU601250816 — REVISION: 1

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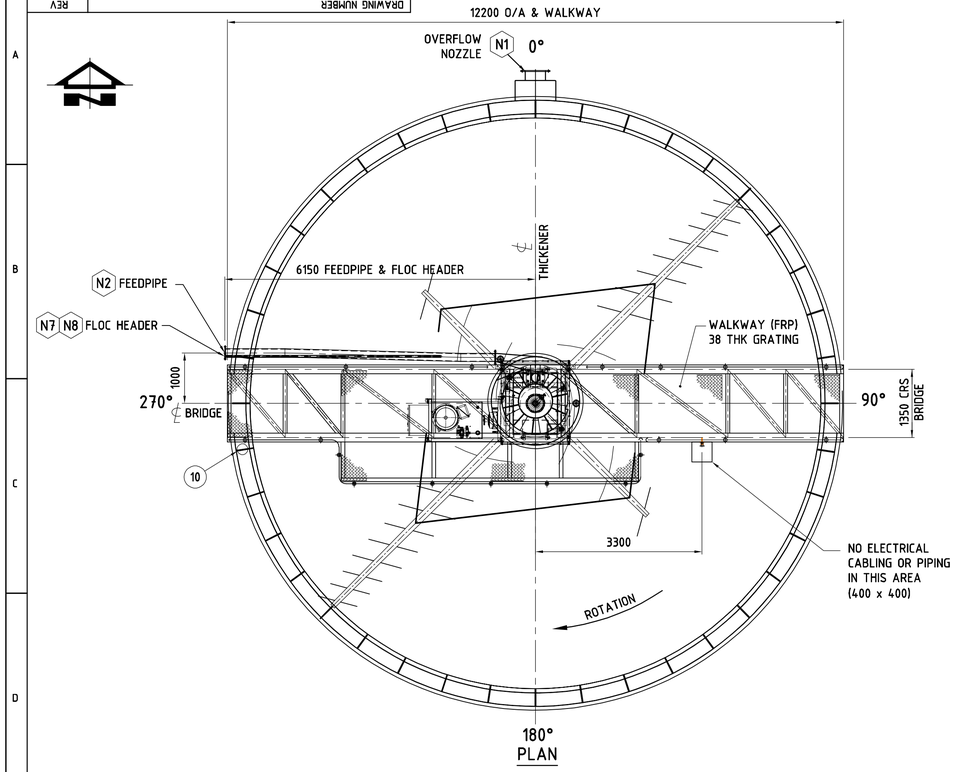
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DRAWN	timkar	13.11.2019
CHECKED	alecha	13.11.2019
ENGINEER	soucha	14.11.2019
APPROVED	ricleh	14.11.2019
EMPLOYER		
SCALE	1:40	A1

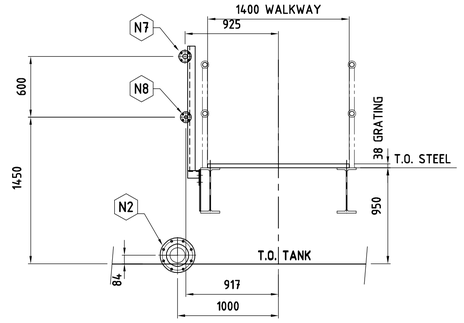
MANSOURAH-MASSARAH GOLD PROJECT
FOOTING AND LOADING PLAN ACIDULATION
3380-THK-001 T40-ACIDULATION THK 12m

DRAWING NUMBER: MD-216-3380-EG-ME-GAD-1002 — REV: A01

DRAWING NUMBER MD - 216 - 3380 - EG - ME - GAD - 1001 REV A01

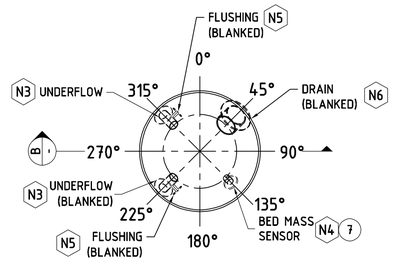


PLAN



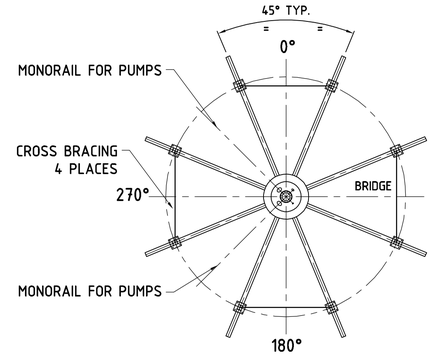
SECTION A-A

SCALE 1:30



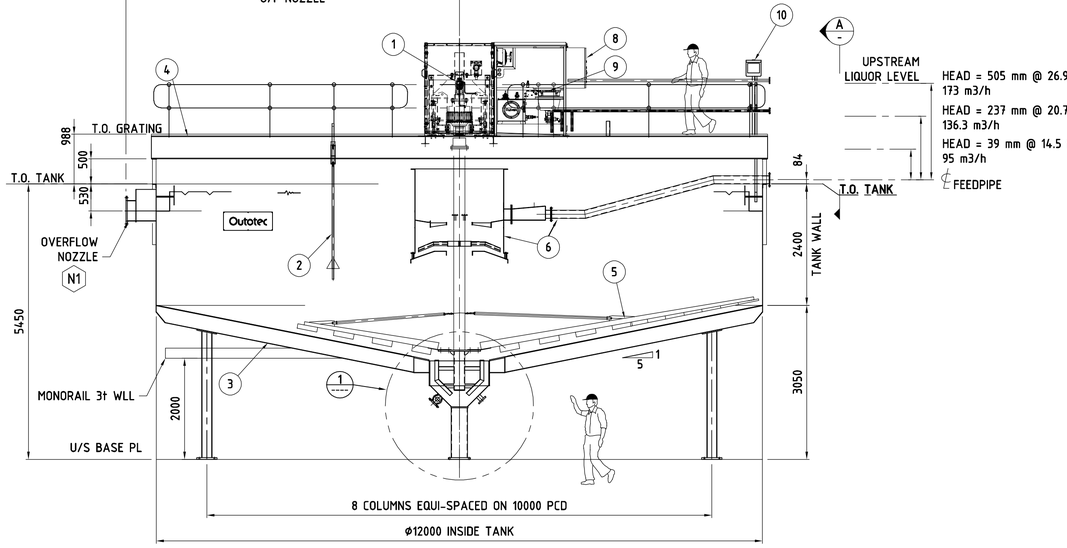
UNDERFLOW CONE DETAILS

SCALE 1:30



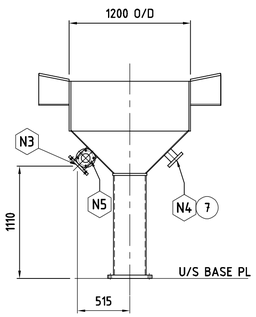
PLAN VIEW
SHOWING TANK SUPPORT ARRANGEMENT

SCALE 1:120

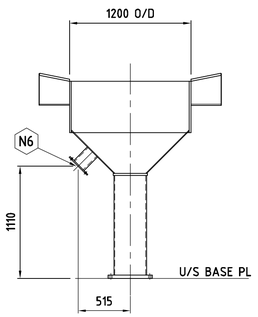


SECTIONAL ELEVATION

REFER TO PLAN FOR TRUE ORIENTATION



DETAIL 1 SCALE 1:30



SECTION B-B

SCALE 1:30

N8	FLOC CONCENTRATE HEADER	DN25	1	1"ASME B16,5-150 lbs
N7	FLOC DILUTION HEADER	DN40	1	1-1/2"ASME B16,5-150 lbs
N6	DRAIN (BLANKED)	DN200	1	8"ASME B16,5-150 lbs
N5	FLUSH (BLANKED)	DN80	2	3"ASME B16,5-150 lbs
N4	BED MASS	DN80	1	3"ASME B16,5-150 lbs
N3	UNDERFLOW (1 DUTY /1 STANDBY)	DN100	2	4"ASME B16,5-150 lbs
N2	FEEDPIPE	DN150	1	12"ASME B16,5-150 lbs
N1	OVERFLOW	DN400	1	16"ASME B16,5-150 lbs
ITEM	DESCRIPTION	SIZE	QTY	CONNECTION TYPE

NOZZLE LIST

10	TURBIDITY SENSOR	-	1	-	-	
9	POWER PACK - 4 kW	-	1	-	-	
8	CONTROL PANEL	-	1	-	-	
7	BED MASS SENSOR	-	1	-	-	
6	FEED ASSEMBLY	MD-216-3380-EG-ME-DED-1004	1	-	-	
5	RAKE MECHANISM ASSEMBLY	MD-216-3380-EG-ME-DED-1003	1	1977	-	
4	BRIDGE ASSEMBLY	-	1	-	-	
3	TANK & SUPPORT STRUCTURE	MD-216-3380-EG-ME-DED-1001	1	15759	-	
2	BED LEVEL SENSOR	-	1	-	-	
1	MEDIUM DRIVE ASSEMBLY - (300 LIFT - REGGIANA RR5200 L4 MS SAI SERIES)	MD-216-3510-EG-ME-DED-1007	1	1470	-	
PART	DESCRIPTION	SUB-DWG.NO.\ MATERIAL	QTY	WEIGHT [KG]	ITEM NO.	REMARKS

NOTE: ACTUAL WEIGHTS TO BE CONFIRMED

SPECIAL MATERIALS AND COATINGS	
UNDERFLOW CONE	CARBON STEEL, SAF2205 CLADDING
UNDERFLOW NOZZLES	SAF2205/316L
TANK SHELL	SAF2205
TANK SUPPORTS	CARBON STEEL PAINTED
MECHANISM	SAF2205
FEEDPIPE	SAF2205
FEEDWELL	SAF2205
FLOC HEADER	316L
WALKWAY GRATING	FRP
ALL WETTED BOLTS	SAF2205

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-ME-DED-1001	GENERAL NOTES								
MD-216-3000-EG-ME-SPC-3001	TECHNICAL SPECIFICATION								
MD-216-3380-EG-ME-DAT-1001	EQUIPMENT DATASHEET	A02	22.11.2019	REVISED AS PER CLIENT COMMENTS	TK	AC	SC	RL	
MD-216-3380-EG-ME-GAD-1002	FOOTING AND LOADING PLAN	A01	04.10.2019	ISSUED FOR APPROVAL	TK	AC	SC	RL	

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LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-G-02853-001 — REVISION: 1

OUTOTEC DRAWING NUMBER: 900707__GAU01__HL15__MTL03__33801 — DOCUMENT ID: 0U601242436 — REVISION: 2

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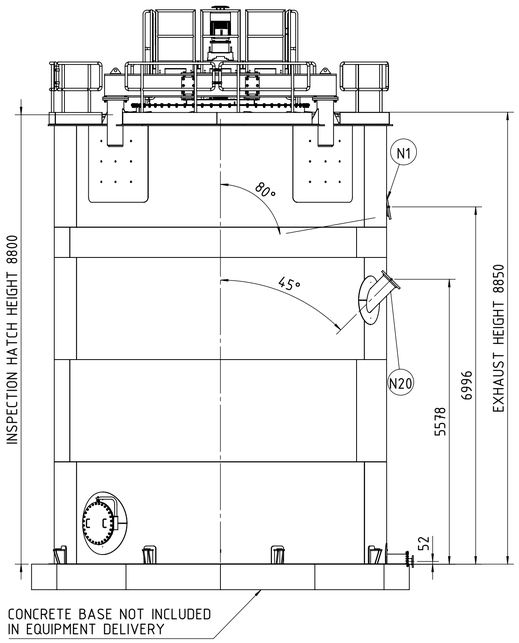
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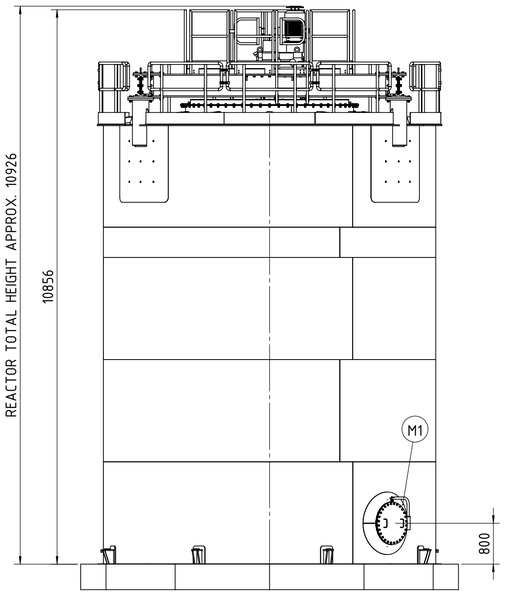
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CHECKED	alecha	20.11.2019	
ENGINEER	soucha	22.11.2019	
APPROVED	ricleh	22.11.2019	
EMPLOYER			
SCALE	1:60	A1	

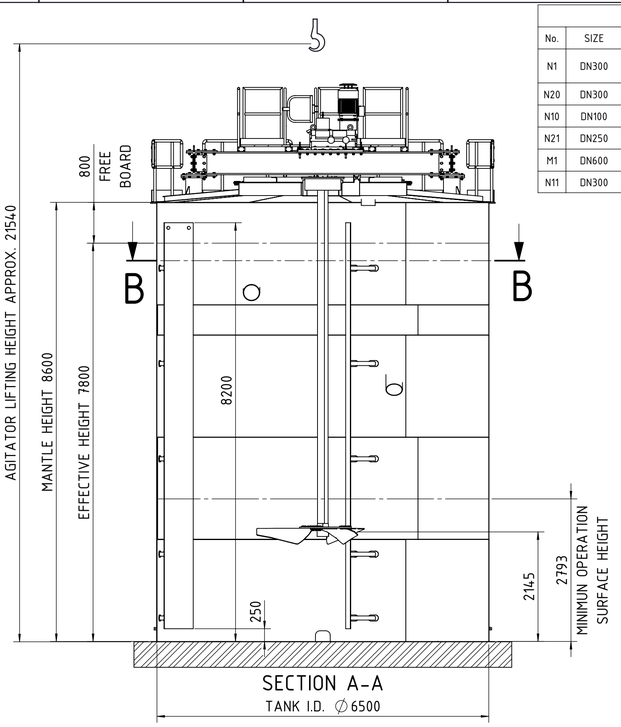
MANSOURAH-MASSARAH GOLD PROJECT
EQUIPMENT GENERAL ARRANGEMENT DRAWING
ACIDULATION
3380-THK-001 T40-ACIDULATION THK, 12m

DRAWING NUMBER: MD - 216 - 3380 - EG - ME - GAD - 1001 — REV: A02

DRAWING NUMBER MD-216-3380-EG-ME-GAD-1013 REV A01



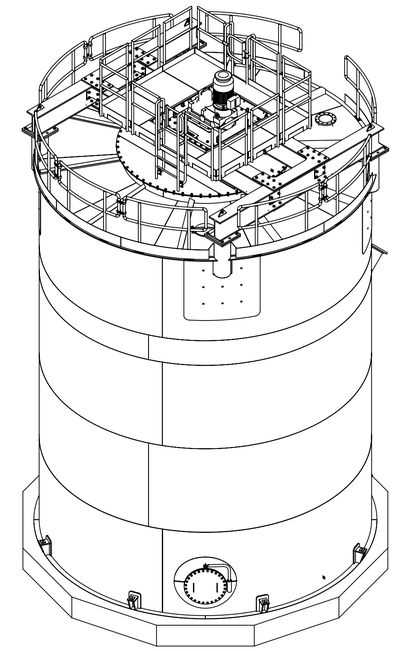


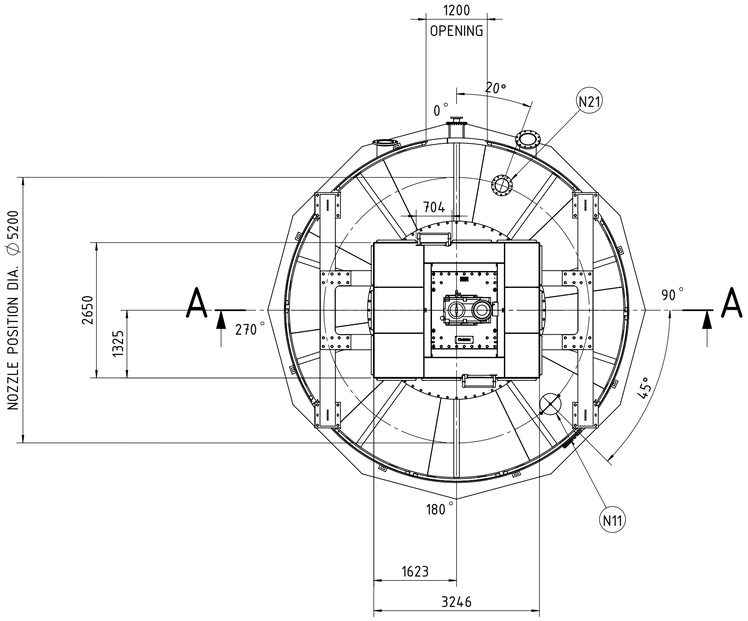


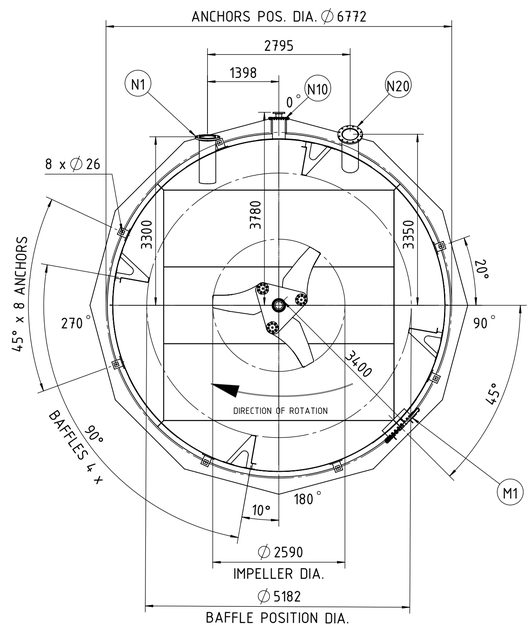
SECTION A-A
TANK I.D. ∅6500

NOZZLE TABLE								
No.	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N1	DN300	CLASS 150	0°	6996	ASME B16.5	INLET NOZZLE	1	
N20	DN300	CLASS 150	0°	5578	ASME B16.5	OUTLET NOZZLE	1	
N10	DN100	CLASS 150	0°	52	ASME B16.5	DRAIN	1	
N21	DN250	CLASS 150	20°	8850	ASME B16.5	EXHAUST	1	
M1	DN600	-	135°	800	OUTOTEC	MANHOLE	1	
N11	DN300	-	135°	8800	OUTOTEC	INSPECTION HATCH	1	

PLEASE CHECK DATASHEET MD-216-3380-EG-ME-DAT-1002 FOR PROCESS VALUES







SECTION B-B

NEUTRALIZATION REACTOR	
AGITATOR WEIGHT APPR. (kg)	3358
DEAD WEIGHT (kg)	22796
OPERATIONAL WEIGHT (kg)	301265
MAXIMUM WEIGHT (kg)	329779

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3380-EG-ME-PLN-1001	FOUNDATION AND LOAD PLAN								
MD-216-3380-EG-ME-DAT-1002	EQUIPMENT DATASHEET								
MD-216-3380-EG-PR-PID-1001	P&I DIAGRAM								
MD-216-3380-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	19.12.2019	ISSUED FOR APPROVAL	MH	TM	JR	GR	

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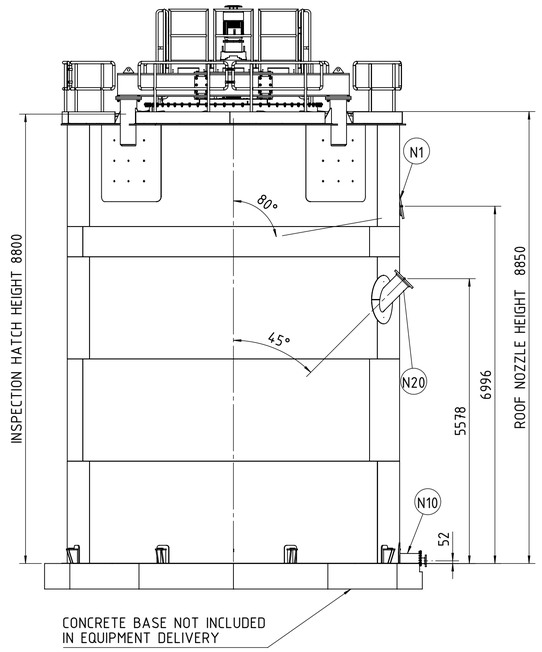
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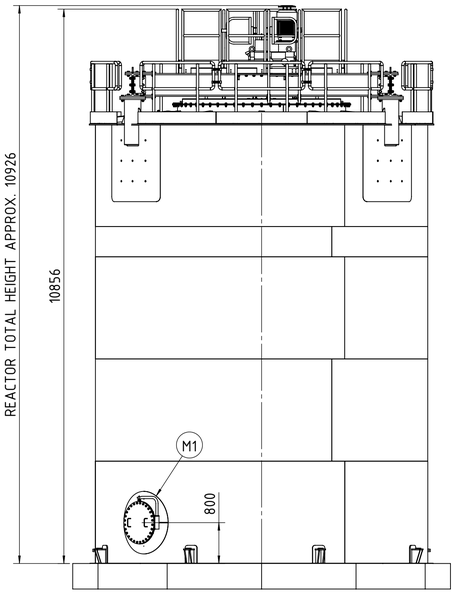
DRAWN	mikhov	18.12.2019
CHECKED	tuoman	19.12.2019
ENGINEER	juhris	19.12.2019
APPROVED	GeoRie	19.12.2019
EMPLOYER		
SCALE	1:60	A1

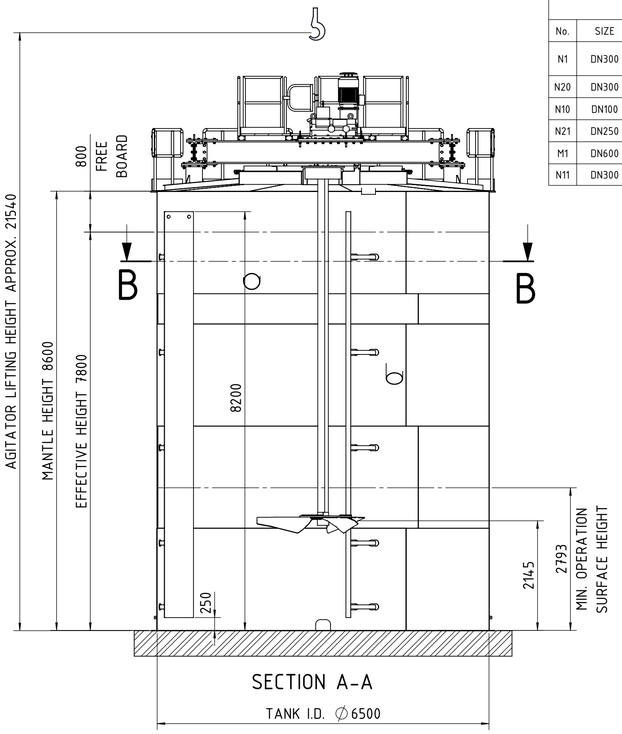
MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
ACIDULATION
3380-REA-001 ACIDULATION REACTOR

DRAWING NUMBER MD-216-3380-EG-ME-GAD-1013 REV A01

DRAWING NUMBER MD-216-3380-EG-ME-GAD-1014 REV A01

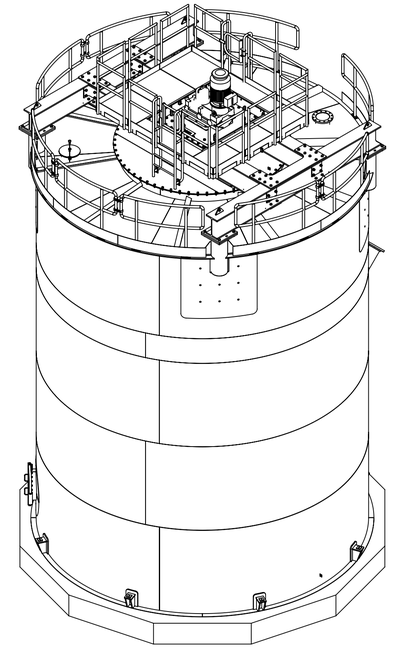


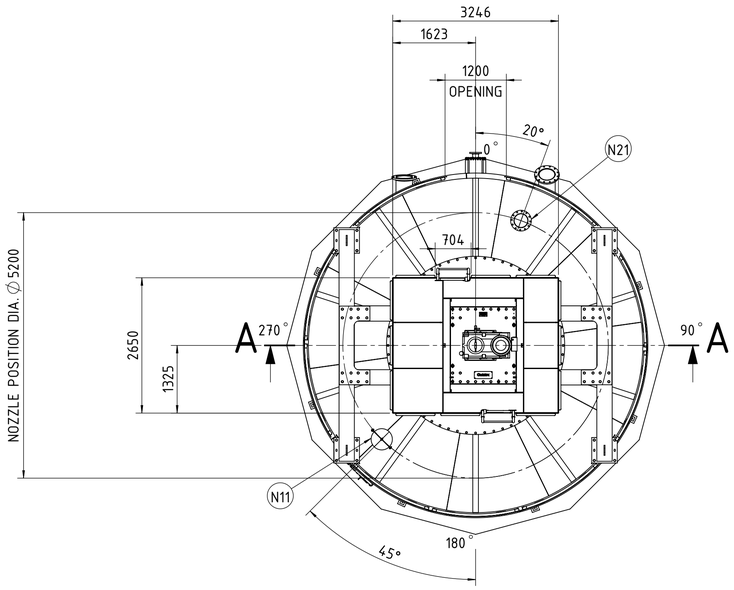


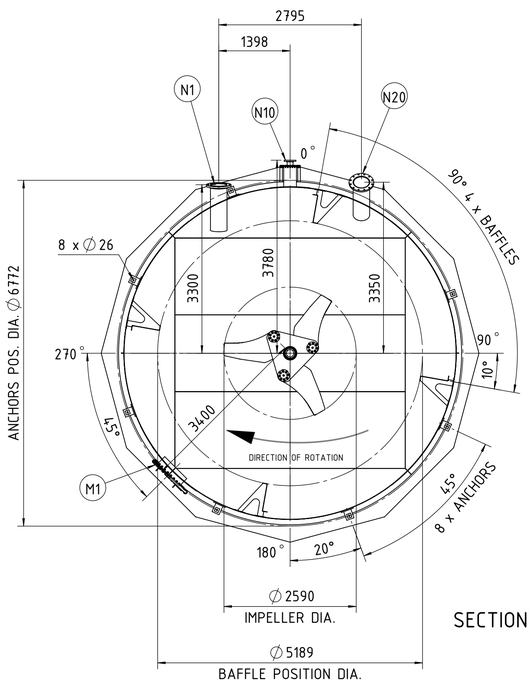


NOZZLE TABLE								
No.	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N1	DN300	CLASS 150	0°	6996	ASME B16.5	INLET NOZZLE	1	
N20	DN300	CLASS 150	0°	5578	ASME B16.5	OUTLET NOZZLE	1	
N10	DN100	CLASS 150	0°	52	ASME B16.5	DRAIN	1	
N21	DN250	CLASS 150	20°	8850	ASME B16.5	EXHAUST	1	
M1	DN600	-	225°	800	OUTOTEC	MANHOLE	1	
N11	DN300	-	225°	8800	OUTOTEC	INSPECTION HATCH	1	

PLEASE CHECK DATASHEET MD-216-3380-EG-ME-DAT-1002 FOR PROCESS VALUES







SECTION B-B

NEUTRALIZATION REACTOR	
AGITATOR WEIGHT APPR. (kg)	3358
DEAD WEIGHT (kg)	22796
OPERATIONAL WEIGHT (kg)	301265
MAXIMUM WEIGHT (kg)	329779

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3380-EG-ME-PLN-1002	FOUNDATION AND LOAD PLAN								
MD-216-3410-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3380-EG-PR-PID-1002	P&I DIAGRAM								
MD-216-3380-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	19.12.2019	ISSUED FOR APPROVAL	MH	TM	JR	GR	

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OUTOTEC DRAWING NUMBER 900707 GAU01 HL15 MTL03 33803 DOCUMENT ID OU601282243 REVISION 0

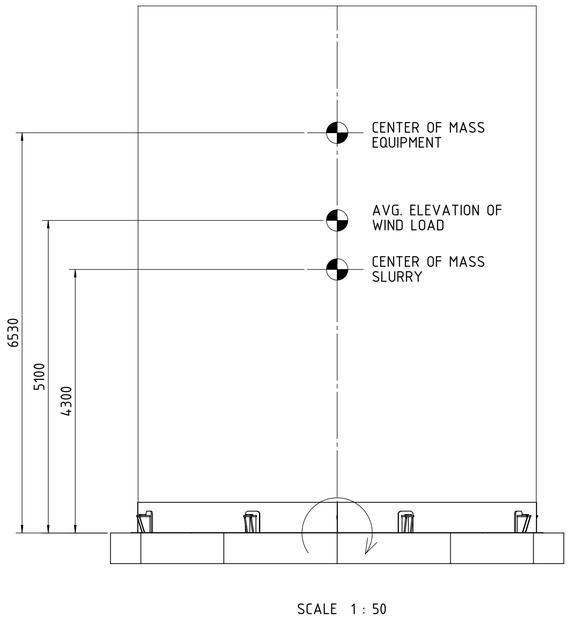
DRAWN	mikhov	18.12.2019
CHECKED	tuoman	19.12.2019
ENGINEER	juhris	19.12.2019
APPROVED	GeoRie	19.12.2019
EMPLOYER		
SCALE	1:60	A1

MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
ACIDULATION
3380-REA-002 ACIDULATION REACTOR

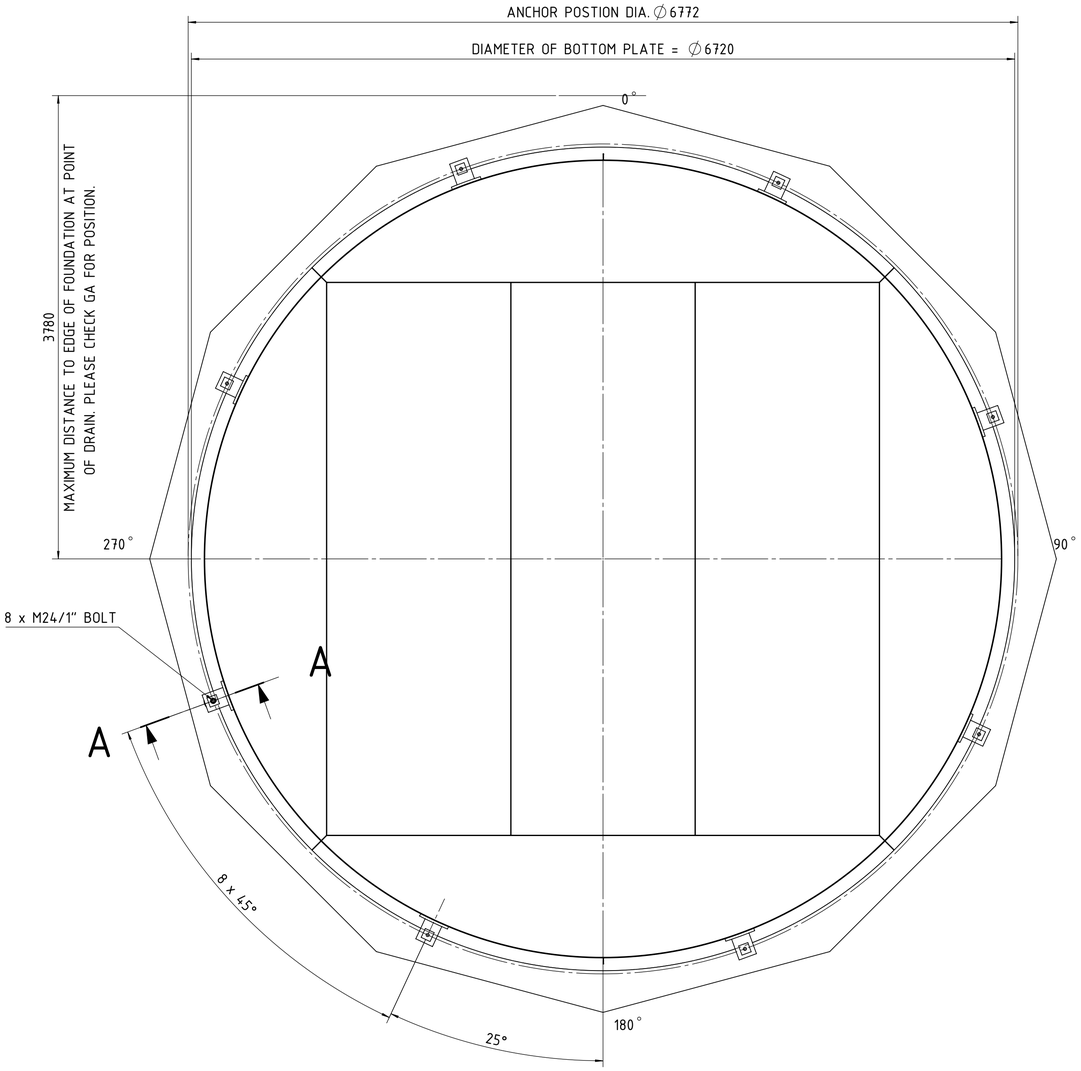
DRAWING NUMBER MD-216-3380-EG-ME-GAD-1014 REV A01

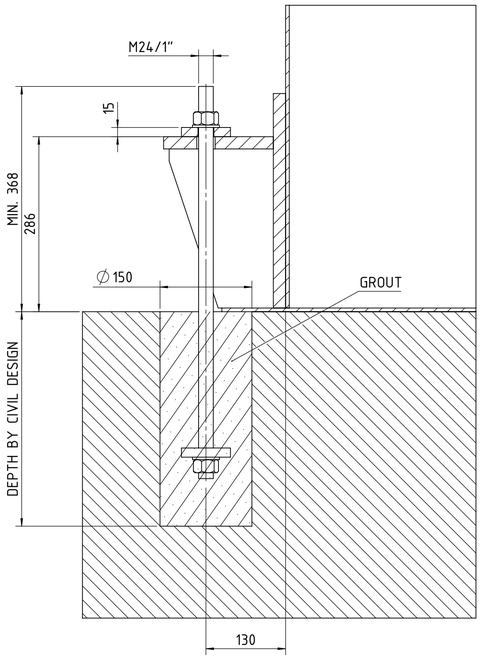
DRAWING NUMBER: MD-216-3380-EG-ME-PLN-1001
REV: A01

LOADING NOTES:	
1	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE CELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SCALE 1 : 50





SECTION A-A
SCALE 1 : 5

ACIDULATION TANK	
DEAD WEIGHT	22800 kg
OPERATIONAL WEIGHT OF SLURRY	278500 kg
MAXIMUM WEIGHT OF SLURRY	307000 kg
EXTERNAL PLATFORM LOAD	8300 kg
NORMAL OPERATING SPEED	44,5 rpm
WIND LOAD	56 kN
SEISMIC LOAD	129 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	575 kNm

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3380-EG-ME-GAD-1001,1002	GENERAL ARRANGEMENT DRAWING
MD-216-3380-EG-ME-DAT-1002	EQUIPMENT DATASHEET
MD-216-3380-EG-PR-PID-1001,1002	P&I DIAGRAM
MD-216-3380-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	20.12.2019	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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DRAWN	tuoman	20.12.2019
CHECKED	aritik	20.12.2019
ENGINEER	juhris	20.12.2019
APPROVED	GeoRie	20.12.2019
EMPLOYER		
SCALE	1:25	A1

MANSOURAH-MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
ACIDULATION
3380-REA-001,002 ACIDULATION REACTOR,

DRAWING NUMBER: MD-216-3380-EG-ME-PLN-1001 — REV: A01

ATTACHMENT 2: COMMENT RESOLUTION LOG SHEET / COMPLIANCE REPORT

PROPRIETARY & CONFIDENTIAL

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT:	**MANSOURAH MASSARAH GOLD PROJECT** **PROJECT NO.: 216**
STAGE:	**4 - PROJECT EXECUTION.** **REVIEW LOG DOC. NO. & REV.:**
ACONEX REF. NO. & DATE:	BECHTEL-WTRAN-001323, 14 February 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 14 February 2020	Comment by	EPC Response DATE: 21 February 2020	Status	Remarks
			General Comments (cc)				
MD-216-3380-EG-CV-CAL-1001-A01	3380	1	Foundations F1 & F2 for the Acidulation Thickener (3380-THK-001), present soil bearing pressures very different between the two foundations. There is a factor of approximately of 2,3 between the average pressures on these foundations. This will produce different settlement between the two foundations and corresponding steel structure. This may affect the steel structure of the thickener. EPC to confirm that this has been taken into account in the design of the steel structure of the thickener	Bechtel / Ma'aden	The dimensions of the foundation marked F2 is increased in order to make the base pressures under the foundations F1 & F2 almost equal. Hence, there shall be no differential settlement between the two foundations which can have an adverse effect on the steel structure for the thickener.	Closed	